

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0685

La Marque 100 kV
July 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0685. The Interconnection Customer proposes a 299.1 MW natural gas combined cycle facility interconnecting at the La Marque 100 kV substation in the Indiana area. The request was received on October 06, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 258 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,254,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: La Marque - Mangum 100 kV Line
Contingency: La Marque - Lagrange Substation 100 kV
Loading: 272.1% (Rating: 104 MVA)
DFAX: 100.0%
Pre-project loading: 16.5%
2. Facility: Lagrange - La Marque 100 kV
Contingency: La Marque - Mangum 100 kV
Loading: 251.4% (Rating: 112 MVA)
DFAX: 100.0%
Pre-project loading: 15.2%
3. Facility: Mangum - Certified 100 kV
Contingency: La Marque - Lagrange 100 kV
Loading: 223.4% (Rating: 119 MVA)
DFAX: 100.0%
Pre-project loading: 28.6%
4. Facility: Idle Hour Substation - Stubey Road 100 kV
Contingency: La Marque - Mngum 100 kV
Loading: 213.7% (Rating: 126 MVA)
DFAX: 100.0%
Pre-project loading: 23.0%
5. Facility: Sturgis - Idle Hour 100 kV
Contingency: Mangum - La Marque 100 kV
Loading: 213.1% (Rating: 125 MVA)
DFAX: 100.0%
Pre-project loading: 25.8%

6. Facility: Kettner - Certified 100 kV
Contingency: La Marque - Lagrange 100 kV
Loading: 208.8% (Rating: 123 MVA)
DFAX: 100.0%
Pre-project loading: 35.2%
7. Facility: Webster Ic - Stubey Rd 100 kV line
Contingency: La Marque - Mangum 100 kV
Loading: 208.8% (Rating: 125 MVA)
DFAX: 100.0%
Pre-project loading: 31.4%
8. Facility: Hales Branch - Ketner 100 kV line
Contingency: La Marque - Lagrange 100 kV
Loading: 194.5% (Rating: 128 MVA)
DFAX: 100.0%
Pre-project loading: 39.8%
9. Facility: Sturgis - Lagrange 100 kV
Contingency: Mangum - La Marque 100 kV
Loading: 186.4% (Rating: 141 MVA)
DFAX: 100.0%
Pre-project loading: 26.0%
10. Facility: Webster Ic - Mears Junction 100 kV
Contingency: Mangum - La Marque 100 kV
Loading: 178.5% (Rating: 140 MVA)
DFAX: 100.0%
Pre-project loading: 35.1%
11. Facility: Constantine Hydro - Mears Junction 100 kV
Contingency: La Marque - Mangum 100 kV
Loading: 153.5% (Rating: 153 MVA)
DFAX: 100.0%
Pre-project loading: 41.5%
12. Facility: Elliott - Hales Branch 100 kV
Contingency: La Marque - Lagrange 100 kV
Loading: 150.3% (Rating: 145 MVA)
DFAX: 100.0%
Pre-project loading: 55.7%
13. Facility: Gates - Elliott 100 kV line
Contingency: La Marque - Lagrange 100 kV
Loading: 123.0% (Rating: 168 MVA)
DFAX: 100.0%
Pre-project loading: 54.8%
14. Facility: Hydro Plant - Three Rivers 100 kV
Contingency: Palisades Substation - Oceanside 500 kV
Loading: 112.9% (Rating: 151 MVA)
DFAX: 5.5%
Pre-project loading: 102.0%
15. Facility: Ridgeline - Hydro Plant 100 kV
Contingency: Palisades - Oceanside 500 kV Line
Loading: 111.3% (Rating: 154 MVA)
DFAX: 5.5%
Pre-project loading: 100.6%
16. Facility: Simpson Stuart Tap - Fishkill Plains 345 kV
Contingency: Hampton Rd - Medora 345 kV

Loading: 106.5% (Rating: 1,517 MVA)

DFAX: 6.0%

Pre-project loading: 105.4%

17. Facility: Hampton Rd - Medora 345 kV

Contingency: Fishkill Plains - Simpson Stuart Tap 345 kV line

Loading: 104.9% (Rating: 1,597 MVA)

DFAX: 6.9%

Pre-project loading: 103.6%

18. Facility: Constantine Hydro Substation - Flowerfield 100 kV

Contingency: La Marque - Mangum 100 kV

Loading: 102.6% (Rating: 130 MVA)

DFAX: 59.9%

Pre-project loading: 35.6%

19. Facility: Stewart Mountain Dam - Ritter Park 345 kV

Contingency: Medora - P.T.S 345 kV

Loading: 101.6% (Rating: 1,471 MVA)

DFAX: 7.0%

Pre-project loading: 100.2%

20. Facility: Ritter Park - Sugar Hill Substation 345 kV

Contingency: Hilltop - Piney Crk Project 500 kV

Loading: 100.7% (Rating: 1,622 MVA)

DFAX: 10.1%

Pre-project loading: 98.9%

21. Facility: Caldwell - Piney Crk Project 345 kV

Contingency: Hilltop - Piney Crk Project 500 kV

Loading: 100.0% (Rating: 1,917 MVA)

DFAX: 8.7%

Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
La Marque - Mangum 100 kV	Rebuild line with higher-capacity conductor	\$3,028,000	100% (\$3,028,000)
Lagrange - La Marque 100 kV	Rebuild line with higher-capacity conductor	\$2,482,000	100% (\$2,482,000)
Mangum - Certified 100 kV	Rebuild line with higher-capacity conductor	\$2,627,000	100% (\$2,627,000)
Idle Hour Substation - Stubey Road 100 kV	Rebuild line with higher-capacity conductor	\$4,287,000	100% (\$4,287,000)
Sturgis - Idle Hour 100 kV	Rebuild line with higher-capacity conductor	\$1,714,000	100% (\$1,714,000)
Kettner - Certified 100 kV	Rebuild line with higher-capacity conductor	\$12,826,000	100% (\$12,826,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Webster Ic - Stubby Rd 100 kV line	Rebuild line with higher-capacity conductor	\$3,641,000	100% (\$3,641,000)
Hales Branch - Ketner 100 kV line	Rebuild line with higher-capacity conductor	\$3,944,000	100% (\$3,944,000)
Lagrange - Sturgis 100 kV	Rebuild line with higher-capacity conductor	\$6,659,000	100% (\$6,659,000)
Webster Ic - Mears Junction 100 kV	Rebuild line with higher-capacity conductor	\$3,933,000	100% (\$3,933,000)
Constantine Hydro - Mears Junction 100 kV	Rebuild line with higher-capacity conductor	\$3,708,000	100% (\$3,708,000)
Elliott - Hales Branch 100 kV	Rebuild line with higher-capacity conductor	\$8,288,000	100% (\$8,288,000)
Gates - Elliott 100 kV line	Rebuild line with higher-capacity conductor	\$2,451,000	100% (\$2,451,000)
Three Rivers - Hydro Plant 100 kV Line	Rebuild line with higher-capacity conductor	\$5,138,000	100% (\$5,138,000)
Ridgeline - Hydro Plant 100 kV	Rebuild line with higher-capacity conductor	\$5,243,000	100% (\$5,243,000)
Simpson Stuart Tap - Fishkill Plains 345 kV	Reconductor line	\$30,154,000	4% (\$1,102,000)
Hampton Rd - Medora 345 kV	Reconductor line	\$10,037,000	4% (\$368,000)
Constantine Hydro Substation - Flowerfield 100 kV	Reconductor line	\$2,402,000	100% (\$2,402,000)
Stewart Mtn Dam - Ritter Park 345 kV	Reconductor line	\$5,027,000	4% (\$185,000)
Ritter Park - Sugar Hill 345 kV	Reconductor line	\$7,785,000	4% (\$288,000)
Caldwell - Piney Creek Project Substation 345 kV	Reconductor line	\$25,403,000	4% (\$940,000)
	Total allocated to project		\$75,254,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.